VPKBIET, Baramati

SY AI&DS BS23207- Advanced Statistics and Numerical Methods

Assignment Unit-6: Numerical Methods

Solve the following.

1. Using Newton's forward difference formula, find the value of $\sqrt{121}$ from the data.

x	120	122	124	126
$y = \sqrt{x}$	10.954	11.045	11.135	11.224

[**Answer:** $\sqrt{121} = 11.000$]

2. Using Newton's backward difference formula, find the value of $\sqrt{155}$ from the data.

x	150	152	154	156
$y = \sqrt{x}$	12.247	12.329	12.410	12.490

[**Answer:** $\sqrt{155} = 12.288$]

3. Using Lagrange's Interpolation Formula, find y at x = 4 from the data.

x	1	2	5
y	1	4	25

[**Answer:** $f(4) = 16$]

4. Approximate $\int_0^2 (x^2 + 1)dx$ using the Trapezoidal Rule with n=4.

[**Answer:** $\int_0^2 (x^2 + 1)dx \approx 4.75$]

5. Evaluate $\int_0^1 \frac{dx}{x^2+1}$ using Simpson's $\frac{1}{3}^{rd}$ rule, (Take h = 0.25).

[**Answer:** $\int_0^1 \frac{dx}{x^2+1} \approx 0.7854$]

6. Approximate $\int_0^3 (x^3 + 1)dx$ using Simpson's $\frac{3}{8}^{th}$ Rule with n=6.

[**Answer:** $\int_0^3 (x^3 + 1)dx \approx 23.25$]

7. Use Runge-Kutta method of fourth order to solve $\frac{dy}{dx} = x + y^2, y(0) = 1$ at $x = 0.1$ with $h = 0.1$.

[Answer: $y_1 = 1.1165$]

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